TEXORA · LECTURE SHEET

Modern Weaving Machineries

The shuttleless revolution: how the projectile, the rapier, the air jet and the water jet each carry a pick across the shed without a shuttle — and why removing the shuttle multiplied the speed, width and quietness of weaving several times over.

TE-2103 · WEAVING TECHNOLOGY · TOPIC 10 · LEVEL 2 TERM 1

1. The shuttleless revolution

For two centuries the pick was carried inside a **shuttle** — a heavy wooden body (300–400 g or more) holding a pirn of weft, thrown from box to box. The shuttle set three hard limits: its mass had to be accelerated and stopped every pick, wasting most of the loom's energy and capping speed at roughly 180–250 picks/min; its size forced a large shed; and its pirn held only a short length of weft, demanding pirn changing and a separate pirn-winding process.

Shuttleless (modern) machines remove the shuttle. Weft is supplied from a large stationary package outside the loom, measured to one pick's length, and carried or blown across the shed by a light carrier — projectile, rapier heads, an air jet or a water jet. The carrier weighs grams, so speeds of 600–1,500+ picks/min become possible, widths grow, noise falls, and pirn winding disappears. The standard comparison measure is the **weft insertion rate**:

$$\text{WIR (m/min)} = \text{picks per minute} \times \text{reed width (m)}$$

WIR combines the two things a shuttle limited — speed and width — into one figure: the total length of weft the machine lays per minute.

WORKED EXAMPLE — WEFT INSERTION RATE, OLD AND NEW

An air-jet loom runs at 1,200 picks/min with a reed width of 1.9 m. A shuttle loom runs at 200 picks/min with a reed width of 1.5 m. Compare their weft insertion rates.

Air-jet: WIR = 1,200 × 1.9 = **2,280 m/min**.

Shuttle loom: WIR = 200 × 1.5 = **300 m/min**.

Ratio = 2,280 ÷ 300 = **7.6** — the air-jet inserts weft 7.6 times as fast. Note that only part of the gain is pick rate (6×); the wider reed adds the rest, which is exactly why WIR, not picks/min alone, is the fair comparison.

2. Projectile weaving machines

Principle. The pick is carried by a small steel **projectile** (about 40–90 mm long, only some 15–40 g). It grips the measured weft in its spring clip and is **fired** across the shed by a torsion-bar picking mechanism releasing its stored energy in a sudden twist, flying through the shed guided by guide teeth that rise into the warp line, dragging the weft behind it. At the receiving side it is caught and braked and the weft released; empty projectiles return underneath the machine on a conveyor chain, several (commonly 4–10 or more) circulating in turn. Selvedges are normally tucked in.

- **Widths and speeds:** single-width to very wide machines — reed widths up to about 3.9 m, and multiple fabric widths side by side on the widest machines; speeds roughly 300–450 picks/min, but WIR is high (up to ≈ 1,400–1,600 m/min) because of the width.
- **Fabric range:** the widest of all systems — light apparel to heavy industrial fabrics, denim, terry, technical textiles; spun and filament wefts, including coarse and fancy yarns, because the projectile grips positively.
- **Character:** positive mechanical insertion, excellent weft control, easy multi-colour work, moderate energy per pick; mechanically complex and capital-heavy.

3. Rapier weaving machines

Principle. The weft is carried by one or two long, thin **rapiers** — tapes or rods with a gripper head. In the common **double rapier** system the **giver rapier** enters from one side carrying the pick and, at the centre of the shed, hands it to the **taker rapier**, which draws it out to the other side as both withdraw. In a **single rapier**, one rapier crosses the whole width, gripping the weft on the far side or carrying it through in a loop. The mid-shed transfer is the heart of the system.

3.1 Rigid and flexible rapiers

FEATURE	RIGID RAPIER	FLEXIBLE RAPIER
Rapier form	Stiff rod or tube, slides straight across; needs a clear straight path and space each side to withdraw — machine footprint wide	Thin steel or composite tape that is guided across the shed and coils into a curved guide or onto a wheel as it withdraws — compact machine
Shed guidance	Head runs free in the shed (no guides inside)	Tape runs through guide hooks / on the race along the shed
Best suited	Very wide widths, heavy and technical fabrics	The general modern standard for apparel and home textiles

- **Widths and speeds:** reed widths commonly 1.5–3.8 m (rigid types wider still); speeds roughly 400–800 picks/min on modern machines; WIR up to ≈ 1,200–1,600 m/min.
- **Fabric range:** the most versatile system of all — fine to coarse, spun or filament, natural, synthetic or blends, single or many weft colours (rapiers are the best at multi-colour and fancy weft insertion, since each colour is positively gripped), delicate yarns handled gently.
- **Character:** positive gripper control from package to selvedge, moderate noise, good for mills weaving many styles in smaller lots.

4. Air-jet weaving machines

Principle. The weft is carried by compressed air. The measured pick is drawn from the supply package by a feeder drum, then the **main nozzle** fires a burst of air launching the yarn into the shed's guide channel — on modern looms formed by the profile reed itself. One jet cannot hold speed across a wide shed, so **relay nozzles** along the reed fire in timed sequence, each refreshing the stream and carrying the pick tip onward to the receiving side, where suction holds the yarn until beat-up. The weft comes **directly from the supply package — there is no pirn** — and nothing mechanical crosses the shed.

- **Widths and speeds:** reed widths commonly 1.5–3.4 m (wider on some models); the fastest of the single-phase systems — roughly 800–1,500+ picks/min depending on width and yarn; WIR up to ≈ 2,500–3,000 m/min.
- **Fabric range:** light to medium fabrics in high volume — shirting, sheeting, lining, lighter denim, filament fabrics; wefts must be strong and smooth enough to fly (weak, hairy or heavy wefts fly badly); weft colours possible but fewer than rapier.
- **Character:** highest production, simplest insertion path — but compressed air is expensive, so air consumption is the main running cost; insertion is not positive, so yarn quality and nozzle timing are critical.

5. Water-jet weaving machines

Principle. A fine jet of water, pumped at high pressure through a single nozzle, carries the pick: the yarn rides inside the coherent water stream across the shed in the same way as in air-jet weaving, but water is denser, so the jet stays compact and powerful over the width with one nozzle and no relays. The machine weaves **wet**: the water is collected under the loom, filtered and recirculated, and the fabric leaves damp and must be dried — the water is **removed after weaving** at a drying stage.

- **The filament-only rule:** restricted to **hydrophobic synthetic filament yarns** (polyester, nylon and similar) that do not absorb water, weaken when wet or swell — natural fibres and spun yarns are excluded.

- **Widths and speeds:** reed widths commonly 1.5–3.3 m; speeds roughly 800–1,200+ picks/min; WIR up to ≈ 2,000–2,600 m/min.
- **Fabric range:** synthetic filament fabrics in bulk — linings, taffeta, dress and umbrella fabrics, some industrial filament cloths.
- **Character:** simple, quiet, low energy per pick (pumping water is cheaper than compressing air); but stainless construction is required and water quality and drainage must be managed.

6. Multiphase weaving, in outline

All the systems above are **single-phase**: one shed, one pick at a time. **Multiphase** machines form several sheds at once and insert several picks simultaneously — in the best-known design a wave of shed openings travels across the machine and small carriers ride the wave, each laying a pick. In principle WIR multiplies by the number of simultaneous picks. In practice selvedge formation, weft control, tension uniformity and fabric range have limited commercial success, and multiphase machines remain a niche: the logical limit of insertion technology, not the present standard.

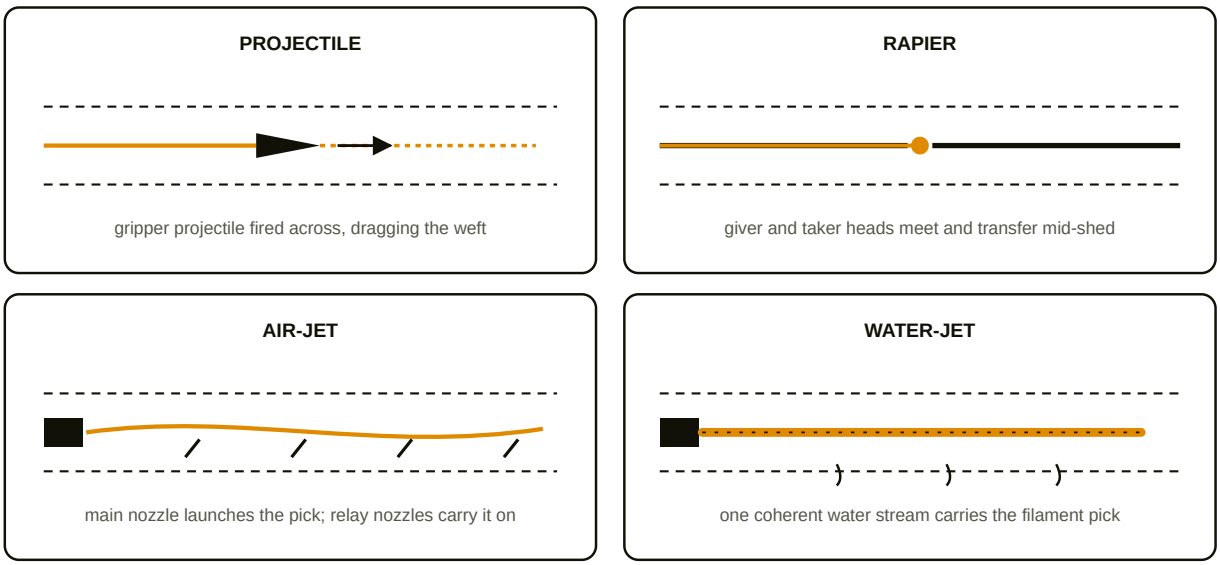


Figure 1 — The four shuttleless insertion principles. Projectile and rapier grip the weft positively and mechanically; air-jet and water-jet carry it in a fluid stream — air needs relay nozzles across a wide shed, the denser water jet does not.

7. The modern machine around the insertion system

Contemporary looms of every insertion type share a layer of electronic equipment that has changed mill life as much as the insertion itself:

- **Electronic shedding** — electronic dobby and jacquard: patterns are files, not lags or cards; changeover is a download.

- **Electronic let-off and take-up** — tension and picks per unit length set digitally and held by servo feedback (Topic 9), constant from full beam to empty and reproducible for repeat orders.
- **Automatic weft repair / restart** — after a weft break the machine finds the shed, removes the faulty pick (automatic pick removal) and restarts itself, cutting weaver intervention and the start-marks that used to follow every stop.
- **Monitoring and data** — sensors watch warp breaks, weft arrival, tension and efficiency; the loom stops itself on faults, displays the cause, and reports production data to the mill network (efficiency per loom, shift and style).
- **Quick style change** — beam, harness and reed exchanged as prepared units, cutting changeover from hours toward minutes.

WORKED EXAMPLE — CLOTH OUTPUT OF AN AIR-JET LOOM

An air-jet loom runs at 1,200 picks/min weaving a fabric of 50 picks/cm. Calculate the cloth output per hour.

Picks needed per metre of cloth = 50 picks/cm × 100 cm = **5,000 picks/m**.

Picks inserted per hour = 1,200 × 60 = **72,000 picks/h**.

Output = 72,000 ÷ 5,000 = **14.4 m of cloth per hour**.

(At a realistic 90 % efficiency this becomes 14.4 × 0.90 ≈ 13.0 m/h — always state whether a figure is theoretical or efficiency-adjusted.)

8. The systems compared

SYSTEM	INSERTION MEDIUM	TYPICAL WIR (M/MIN)	TYPICAL REED WIDTH	SUITABLE YARNS / FABRICS	NOISE / ENERGY NOTES
Shuttle (for reference)	Shuttle carrying a pirn	≈ 200–400	1.0–1.9 m typical	Almost anything, slowly	Loudest (picking + box impact); energy wasted moving the heavy shuttle
Projectile	Small gripper projectile, fired by torsion bar	Up to ≈ 1,400–1,600	Up to ≈ 3.9 m, multi-width possible	Spun and filament, coarse to fine; wide and heavy fabrics, technical textiles	Moderate noise; energy only to move a light projectile — efficient per pick

Rapier	One or two gripper rapiers (rigid or flexible)	Up to ≈ 1,200–1,600	≈ 1.5–3.8 m (rigid wider)	Widest versatility: all yarn types, many weft colours, fancy and delicate wefts	Moderate noise; moderate energy
Air-jet	Compressed-air stream: main + relay nozzles	Up to ≈ 2,500–3,000	≈ 1.5–3.4 m	Light–medium fabrics; smooth, strong wefts; high-volume plain fabrics and filament	Air hiss; highest energy cost — compressing air is expensive
Water-jet	Single high-pressure water jet	Up to ≈ 2,000–2,600	≈ 1.5–3.3 m	Hydrophobic synthetic filament only; fabric woven wet, dried after	Quiet; low energy per pick; water handling and corrosion-proof build required
Multiphase	Several carriers, several sheds at once	Highest in principle	Special designs	Limited fabric range in practice	Complex; niche commercial use

Choosing a system is choosing the compromise a mill can live with: widest and heaviest fabrics point to projectile; style variety and colour point to rapier; maximum volume of light fabric points to air-jet; bulk synthetic filament points to water-jet. No system is best at everything — versatility, speed, width and running cost trade against each other in every purchase decision.

Glossary

Shuttleless weaving — pick inserted by a light carrier or fluid jet instead of a shuttle, weft coming from a stationary package. **Weft insertion rate (WIR)** — picks per minute × reed width in metres; the standard measure of insertion performance. **Projectile** — the small gripper element fired across the shed carrying the weft. **Rapier** — rod or tape with a gripper head carrying the weft; giver and taker transfer the pick at mid-shed. **Main nozzle / relay nozzles** — the nozzle that launches the pick, and the sequence along the reed sustaining the air stream. **Profile reed** — the shaped reed whose channel guides air stream and pick in air-jet weaving. **Multiphase weaving** — several sheds formed and several picks inserted simultaneously. **Automatic pick removal** — extracting a faulty pick after a weft break so weaving restarts without a mark.

Key points

• Removing the shuttle removed the speed, width and energy limits it imposed: modern carriers weigh grams, weft comes straight from a supply package, and no pirn is needed.
• WIR = picks/min × reed width in metres — the fair way to compare machines of different widths.
• Projectile: positive grip, widest widths, heavy and technical fabrics. Rapier: giver–taker transfer mid-shed, rigid or flexible, single or double — the most versatile system, best for colours and fancy wefts.
• Air-jet: main nozzle plus timed relay nozzles, fastest single-phase system, but air is costly and insertion is not positive. Water-jet: one coherent jet, synthetic filament only, cloth woven wet and dried after.
• Multiphase inserts several picks at once — highest WIR in principle, limited in practice by selvedge, control and fabric-range problems.
• Around every modern insertion system sit electronic dobby/jacquard, electronic let-off and take-up, automatic weft repair and full machine monitoring.

Practice

1. A rapier loom runs at 650 picks/min with a reed width of 2.3 m. Calculate its weft insertion rate.
2. A water-jet loom runs at 1,000 picks/min weaving filament fabric of 40 picks/cm. Calculate the theoretical cloth output in metres per hour.
3. Calculate the WIR of (a) a projectile loom at 350 picks/min and 3.6 m reed width and (b) an air-jet loom at 900 picks/min and 1.8 m reed width. Which inserts weft faster, and by what percentage?
4. A mill must weave (i) a 3.4 m wide heavy industrial fabric, (ii) a fashion fabric using six weft colours in small lots, and (iii) bulk polyester filament lining. Choose an insertion system for each and justify each choice in one sentence.
5. Explain why water-jet weaving is impossible for cotton spun yarn, covering what the water does to the yarn and to the fabric.
6. State three things the electronic equipment of a modern loom does besides inserting the weft, and the mill-level benefit of each.

Answer key

1. WIR = 650 × 2.3 = **1,495 m/min**.
2. Picks per metre = 40 × 100 = 4,000 picks/m. Picks per hour = 1,000 × 60 = 60,000. Output = 60,000 ÷ 4,000 = **15 m/h** (theoretical, before efficiency).
3. (a) 350 × 3.6 = **1,260 m/min**. (b) 900 × 1.8 = **1,620 m/min**. The air-jet inserts faster. Difference = 1,620 − 1,260 = 360 m/min; 360 ÷ 1,260 × 100 = **28.6 % faster** — despite the projectile loom being twice as wide, the air-jet's pick rate more than compensates.

4. (i) **Projectile** — only it combines very wide width with a positive grip strong enough for heavy industrial weft. (ii) **Rapier** — positive gripper control of each colour makes it the best multi-colour, small-lot, style-changing system. (iii) **Water-jet** — built exactly for hydrophobic synthetic filament in bulk, at low energy cost, and polyester filament is unaffected by the water.
5. Cotton is hydrophilic: the yarn absorbs the jet's water, gaining weight and losing clean flight, while the stream breaks up against a rough, hairy spun yarn and cannot carry it reliably. The fabric would leave the loom soaked, swollen and marked, demanding heavy drying. The process works only because synthetic filaments neither absorb water nor weaken when wet.
6. Any three: electronic dobby/jacquard — pattern changed by file download, so style changeovers take minutes and patterns are stored with the order; electronic let-off/take-up — tension and pick density held automatically and reproduced exactly on repeat orders; automatic pick removal and restart — fewer weaver interventions and no start-marks after weft breaks; monitoring and production data — faults stop the machine with the cause displayed, and efficiency by loom/shift/style is visible to management.

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